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PROJECT REPORT

HIDDEN MARKOV MODELS FOR BRAIN DYNAMICS

GROUP 08

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Despite our best efforts, due to our limited experience in both computational modeling and clinical applications, this report may still contain shortcomings. We greatly appreciate any constructive feedback and suggestions from our supervisors and readers to help us improve our work and learn valuable lessons for future projects.

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1 Introduction

1.1 Research Context

The human brain is not a static entity but a highly complex dynamic network that continuously transitions between distinct functional states on a millisecond timescale. Identifying these microstates and their transition dynamics is crucial for understanding cognitive processes and underlying neural mechanisms.

To observe these rapid temporal changes, Electroencephalography (EEG) remains one of the most widely used electrophysiological modalities due to its exceptional temporal resolution. However, exploiting EEG data presents significant challenges. Signals recorded at the scalp (sensor-level) suffer from severe spatial degradation caused by volume conduction. The superimposition of multiple neural sources, combined with biomechanical artifacts, makes the disentanglement and interpretation of brain states a highly complex data analysis problem.

1.2 Problem Statement

For decades, traditional EEG analysis has relied on time-averaging methods, most notably Event-Related Potentials (ERPs). While useful for observing macroscopic responses (e.g., the P300 wave), ERP inherently suppresses the continuous oscillatory nature and the richness of transient neural networks.

Recently, Hidden Markov Models (HMMs) have emerged as a principled and groundbreaking approach. Leveraging unsupervised learning, HMMs can automatically discover hidden brain states from electrophysiological time series without relying on task labels. Although HMMs have achieved notable success in Magnetoencephalography (MEG), applying this model directly to multi-channel sensor-level EEG introduces substantial hurdles. Specifically, traditional feature extraction methods like the Amplitude Envelope often produce non-Gaussian data distributions with highly correlated spatial covariance matrices. This directly violates the core mathematical assumptions of the HMM, leading to catastrophic state collapse and heavily skewed analytical outcomes.

1.3 Objectives and Contributions

This study aims to optimize the modeling of brain dynamics using high-density EEG data. The core objective is to evaluate how signal preparation methods affect HMM convergence and the accuracy of cognitive decoding.

To achieve this, we constructed a comprehensive signal processing pipeline using the *OSL (Oxford Software Library)* toolbox. This pipeline involves source reconstruction from sensor-level EEG to mitigate spatial leakage, while integrating a custom automated data trimming solution (Auto-Trimmer) to overcome stringent computational resource limits (e.g., RAM constraints).

The primary contribution of this research is a comparative empirical perspective on the Nencki-Symfonia dataset (Auditory Oddball task). We implemented and directly contrasted two state-of-the-art HMM architectures:

- **AE-HMM (Amplitude Envelope HMM):** A model based on signal power envelope extracted via the Hilbert transform.

- **TDE-HMM (Time-Delay Embedded HMM):** A model that preserves the integrity of spectral and phase information through time-delay embedding combined with dimensionality reduction (PCA).

1.4 Research Questions

The experimental design and analysis in this report are structured to address two primary Research Questions (RQs):

RQ1: *How do signal quality and preparation methods (particularly the preservation of spectral information and distributional structure) affect the association between HMM state parameters and cognitive processes in an EEG Oddball task?*

RQ2: *Between the AE-HMM and TDE-HMM variants, which method provides superior dynamical features that maximize classification accuracy when distinguishing between Standard and Target stimuli?*

2 Theoretical Background

2.1 Brain Dynamics

The brain operates as a complex, non-linear network system where functional regions continuously interact to process cognitive information. Brain dynamics refer to how these coordinated patterns evolve over extremely short timescales. Rather than maintaining a static state, the brain continuously fluctuates between temporarily stable microstates. These states represent the spontaneous or task-oriented organization of large-scale functional networks. Accurately modeling these dynamic transitions is key to decoding how the brain responds to external environmental stimuli.

2.2 Electroencephalography (EEG) and the Oddball Paradigm

2.2.1 Physiological Nature of EEG Signals

EEG records the electrical potentials generated by the accumulated current from neural activity, specifically the synchronized postsynaptic potentials of pyramidal cells in the cerebral cortex. EEG signals provide millisecond-level temporal resolution, allowing direct observation of real-time cognitive dynamics. However, as the signals propagate through the meninges, skull, and scalp, volume conduction occurs, causing spatial blurring and high correlations among adjacent sensors.

2.2.2 Oddball Paradigm and the P300 Marker

The Auditory Oddball Paradigm is a classic experimental model in cognitive science used to study attention and information processing. In this protocol, a sequence of auditory stimuli is presented, comprising two main types: Standard stimuli, which appear with high frequency (approx. 85% of trials), and Target stimuli, which appear randomly and infrequently (approx. 15%), requiring the participant to execute a specific behavioral response (e.g., a button press).

Neurophysiologically, Target stimuli tend to elicit a specific Event-Related Potential (ERP) known as the P300 wave. The P300 is a positive voltage deflection that typically peaks between 250 and 450 ms post-stimulus, concentrated heavily in the centro-parietal region. The amplitude and latency of the P300 directly reflect the allocation of attentional resources, detection of environmental changes, and working memory updating.

2.3 Hidden Markov Models (HMM)

2.3.1 General Mathematical Mechanism

The Hidden Markov Model (HMM) provides an ideal probabilistic framework for describing neurophysiological time series. An HMM assumes that the observed signal sequence $Y = \{y_1, y_2, \dots, y_T\}$ is stochastically generated by a sequence of unobservable hidden states $S = \{s_1, s_2, \dots, s_T\}$, where each hidden state s_t belongs to a discrete set of K states $\{1, 2, \dots, K\}$.

The transition between hidden states follows a first-order Markov property, meaning the probability of transitioning to the next state depends solely on the current state:

$$P(s_t = j \mid s_{t-1} = i, s_{t-2} = k, \dots) = P(s_t = j \mid s_{t-1} = i) = A_{ij} \quad (1)$$

where $A = \{A_{ij}\}$ is a $K \times K$ transition probability matrix. At any time t , if the system is in hidden state k , the observed signal y_t is emitted from a continuous probability distribution (Emission Distribution), commonly modeled as a Multivariate Gaussian Distribution:

$$y_t \mid s_t = k \sim \mathcal{N}(\mu_k, \Sigma_k) \quad (2)$$

where μ_k is the mean vector and Σ_k is the covariance matrix characterizing the spatial activity of state k .

2.3.2 Amplitude Envelope HMM (AE-HMM)

The AE-HMM focuses on modeling the power dynamics of the signal. Prior to HMM training, band-limited signals are processed via the Hilbert transform to extract the instantaneous amplitude envelope, which is then smoothed using a moving average filter.

While effective in source-space MEG analysis, AE-HMM exhibits a fundamental weakness when applied to multi-channel sensor-level EEG. The nature of Hilbert-transformed amplitude envelope data is strictly non-negative and heavily right-skewed, directly violating the multivariate Gaussian assumption of the HMM emission model. Forcing this non-Gaussian data into the model often causes the optimization process to become trapped in poor local minima, leading to state collapse—a phenomenon where the model assigns all data to a single state with massive variance to mathematically minimize the loss function, thereby losing all physiological relevance.

2.3.3 Time-Delay Embedded HMM (TDE-HMM)

To overcome the limitations of AE-HMM, the TDE-HMM preserves both the amplitude and phase information of the oscillatory signal. For each data channel, the model automatically augments the signal with time-shifted copies (lags) of itself within a specified time window (e.g., ± 7 samples around the current time point).

This time-delay embedding expands the vector space, allowing the HMM to directly model the spectral structure and cross-correlations between brain regions. Because raw EEG signals are approximately symmetric with a near-zero mean, the TDE-embedded data maintains a near-perfect Gaussian distribution, strictly satisfying the mathematical assumptions of the HMM emission model. As dimensionality spikes following embedding (Original Channels $\times$ Number of Lags), Principal Component Analysis (PCA) is subsequently applied to decorrelate the spatial dimensions and compress the data into a core set of components (e.g., 80 components), stabilizing the covariance matrices and optimizing computational performance.

2.4 Overview of the OSL (Oxford Software Library) Toolbox

The OSL (Oxford Software Library) is an open-source ecosystem built on Python, specifically designed for advanced analysis of electrophysiological data (EEG/MEG). This ecosystem is divided into two independent yet closely coordinated core libraries:

- **osl-ephys:** Handles the entire raw sensor-level data preprocessing pipeline, spatial channel standardization, geometrical coregistration via the RHINO module, and Source Reconstruction using LCMV Beamforming coupled with orthogonalization to project scalp signals into specific intracranial brain regions.
- **osl-dynamics:** Focuses on building and training probabilistic machine learning models for brain dynamics. This library provides high-performance, GPU-optimized APIs to prepare large structural data (TDE, PCA), train TDE-HMM models using Variational Free Energy optimization, and offers post-processing tools to decode real-time state sequences (Viterbi path extraction) as well as compute functional network connectivity matrices.

3 Experimental Setup and Pipeline

3.1 Dataset

This study utilizes data from the publicly available Nencki-Symfonia EEG dataset. The original dataset provides high-density EEG recordings from 42 healthy young adults performing four distinct cognitive tasks. For the scope of this study, we focused exclusively on the dataset associated with the *Auditory Three-stimulus Oddball* task.

The protocol for this task consists of three trial types corresponding to different auditory stimuli:

- **Standard Stimulus (Label S5):** A frequent, familiar "p" tone (accounting for approximately 85% of the total trials).
- **Target Stimulus (Label S6):** A rare "+" tone requiring participants to pay attention and execute a behavioral response (approx. 15%).
- **Distractor Stimulus (Label S7):** Another rare, deviant tone (approx. 15%) designed to test attention dispersion.

Technically, electrophysiological recordings were acquired using a dedicated BrainVision system configured with a 128-channel EEG setup. All data were collected at the sensor-level with a native sampling rate of 1000 Hz, ensuring maximum temporal resolution for subsequent dynamical analyses.

3.2 Preprocessing and Source Reconstruction

Due to memory and time constraints, the experiment was limited to 30 subjects. The processing and source reconstruction pipeline is illustrated below:

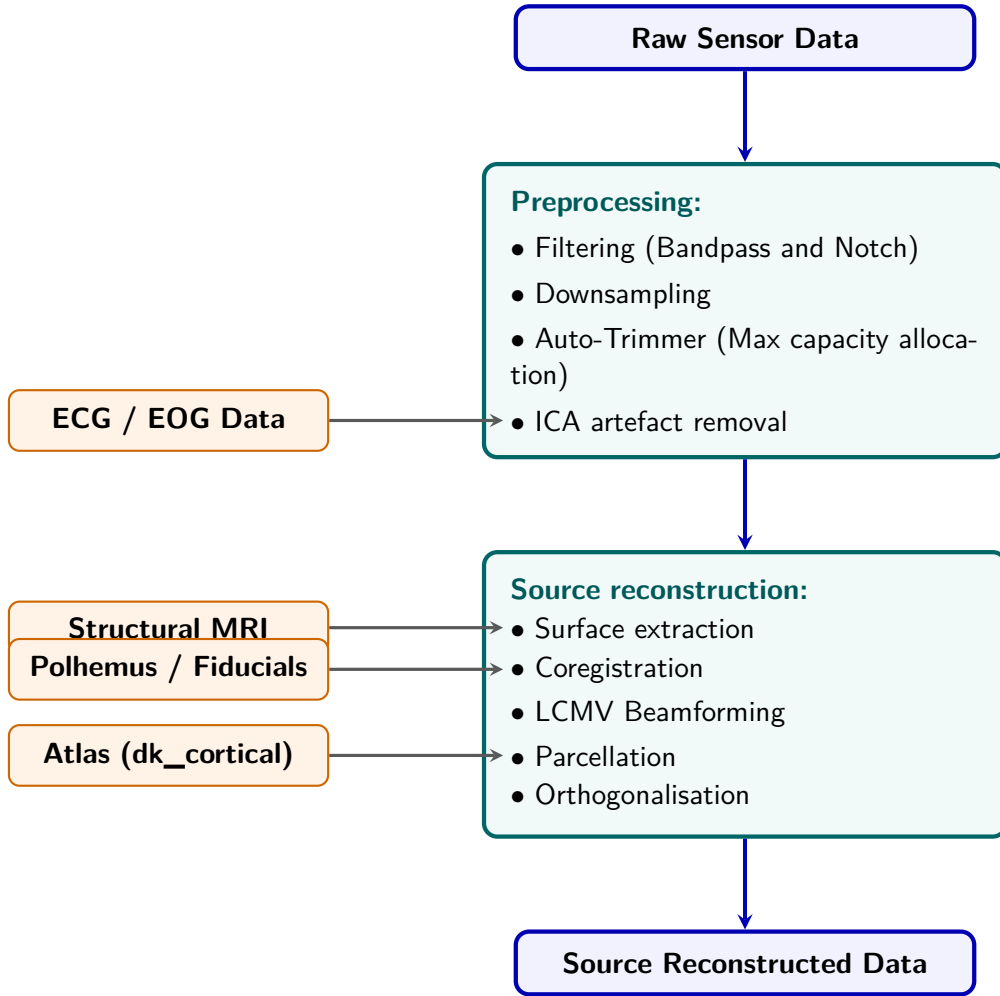


Figure 1: Schematic overview of the signal preprocessing and source reconstruction pipeline using the `osl-ephys` library.

3.2.1 Signal Preprocessing

The data cleaning process was automated using the `osl-ephys` library. Initially, the signal was passed through a bandpass IIR filter (0.5 to 45 Hz) to eliminate slow drifts and high-frequency noise unrelated to cognitive processes. A notch filter at 50 Hz was also applied to suppress line noise.

To resolve computational resource limitations, we designed and integrated a custom capacity control algorithm (*Auto-Trimmer*). This algorithm automatically truncates the

first minute of data (often containing artifacts from patient preparation movements) and strictly limits the maximum analysis length of each record to 15 minutes. Subsequently, the data were downsampled from 1000 Hz to an optimal rate to conserve memory while preserving all necessary frequency bands.

3.2.2 Source Reconstruction

To overcome the spatial resolution constraints of sensor-level data, the signals were projected back into the intracranial source space using the RHINO pipeline in `osl-ephys`. Because the raw data lacked digitized head point coordinates, we applied a Virtual Fiducials technique based on standard anatomical landmarks (Nasion, LPA, RPA) to force the system to coregister with the standard MNI152_T1_2mm MRI template.

Next, the Linearly Constrained Minimum Variance (LCMV) Beamforming algorithm was employed to estimate neural current intensities at the source locations. These signals were then parcellated into 68 Regions of Interest (ROIs) based on the `dk_cortical` atlas. Finally, Symmetric Orthogonalisation was applied to minimize spatial leakage, ensuring signal independence across ROIs.

3.2.3 Source Reconstruction Outcomes

Following data preprocessing and source reconstruction, we visualized group-average quantitative metrics confirming that the source-level signals achieved high neurophysiological reliability, establishing a robust foundation for the HMM:

- **Power Spectral Density (PSD):** The PSD plot (Figure 2) strictly adheres to the $1/f$ energy decay law and displays a highly distinct Alpha peak (7-13 Hz), proving that core physiological oscillations were preserved intact.

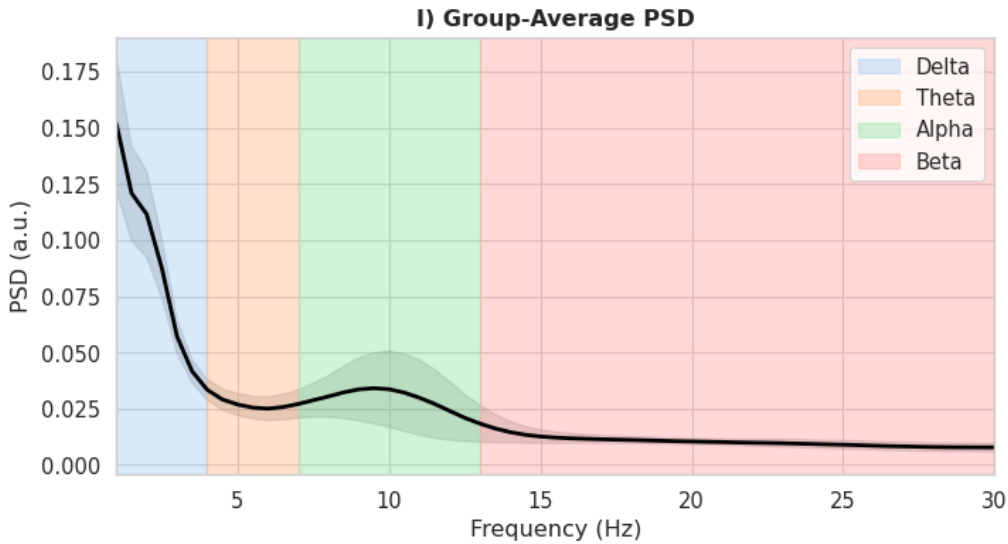


Figure 2: Group-Average Power Spectral Density (PSD) in the source space.

- **Power Maps:** The spatial energy distribution (Figure 3) perfectly aligns with neurophysiological literature: the Alpha band is highly concentrated in the occipito-parietal region, while Delta and Theta bands are spread across the frontal lobes.

II) Group-Average Power Maps Across Frequency Bands

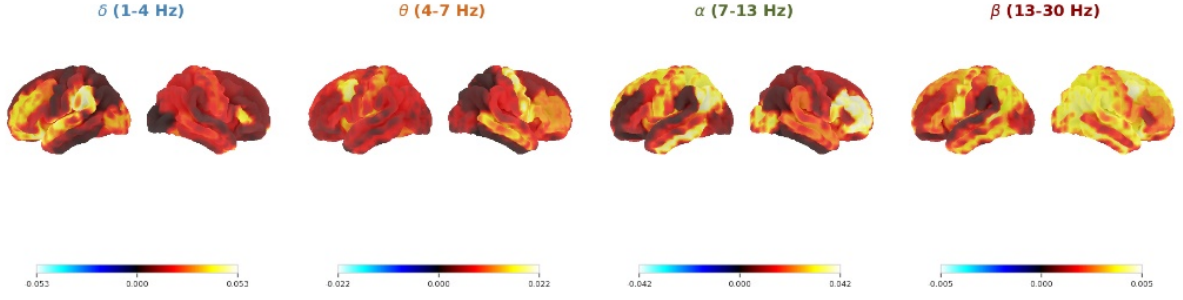


Figure 3: Group-Average Power Maps across Delta, Theta, Alpha, and Beta frequency bands.

- **Connectivity Networks:** Both Phase Synchronization (Coherence - Figure 4) and Amplitude Envelope Correlation (AEC - Figure 5) metrics reveal dense connectivity networks, particularly along the anterior-posterior and interhemispheric axes. This provides an ideal multidimensional feature space for the HMM to segregate states.

III) Group-Average Coherence Networks

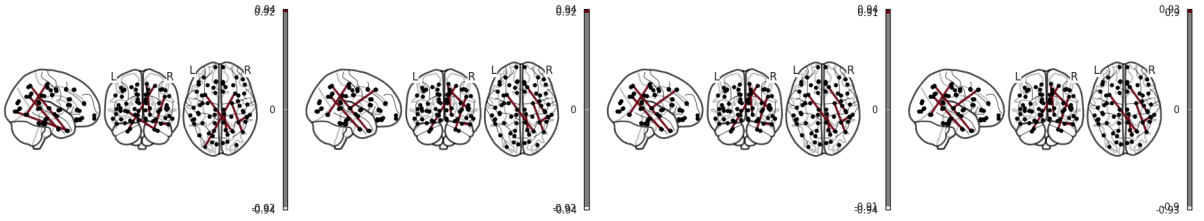


Figure 4: Group-Average Coherence Networks.

IV) Group-Average AEC Networks

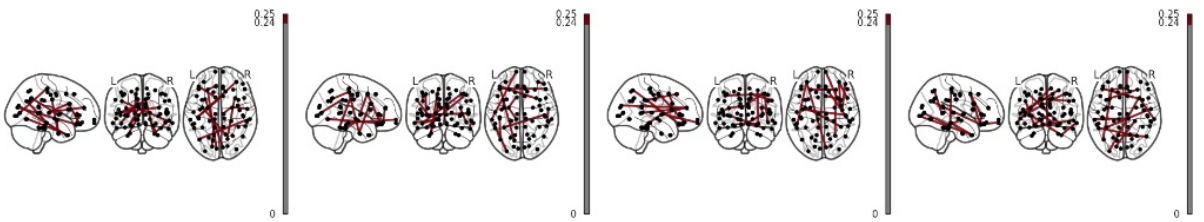


Figure 5: Group-Average Amplitude Envelope Correlation (AEC) Networks.

3.3 HMM Training

We trained two variants of the HMM using the `osl-dynamics` library. Both models were configured to search for $K = 8$ hidden states, assuming the data follows a multi-variate Gaussian emission distribution.

3.3.1 Data Preparation for AE-HMM

The input data preparation pipeline for the Amplitude Envelope HMM (AE-HMM) included the following steps:

1. Bandpass filtering the signal from 1–45 Hz.
2. Applying the Hilbert transform to extract the amplitude envelope.
3. Smoothing the data using a moving average with a window size of 5 samples (equivalent to 20 ms).
4. Applying per-channel standardization.
5. Reducing dimensionality via PCA (compressing 123 raw channels down to 30 components, combined with data whitening).

The PCA dimensionality reduction step was introduced after initial training attempts on all 123 channels encountered numerical instability (returning NaN loss values).

3.3.2 Data Preparation for TDE-HMM

For the Time-Delay Embedded HMM (TDE-HMM), the data processing pipeline varied slightly:

1. Performing time-delay embedding with 15 lags (extracting ± 7 samples around each time point).
2. Reducing dimensionality via PCA (compressing the $123 \times 15 = 1845$ dimensional matrix down to 80 components, including whitening).
3. Applying per-channel standardization.

The primary advantage of TDE-HMM over AE-HMM is its ability to preserve both amplitude and phase information. Consequently, the generated data structure closely approximates a near-Gaussian distribution, making it highly suitable for the HMM observation model.

3.3.3 Training Configuration

Both models were trained in a computing environment utilizing an NVIDIA P100 GPU (16GB VRAM). The training process was distinctly divided into two phases:

- **Phase 1 - Initialization:** The system performed multiple random initializations ($N_{init} = 10$). In each initialization, the model was rapidly trained for 5 epochs on a subset (25% random data). The initialization yielding the lowest negative log-likelihood was selected as the official starting point.
- **Phase 2 - Full Training:** The best weights obtained were used to train the entire dataset for 20 epochs. The training parameters were set as follows: sequence length of 1000 samples (equivalent to 4 seconds), batch size of 16, learning rate of 10^{-3} , Adam optimizer, and covariance regularization parameter $\epsilon = 10^{-6}$. Means learning was enabled for AE-HMM but disabled for TDE-HMM.

Upon completion of the training, we obtained the state probabilities (denoted as α) for each subject at every time point:

$$\alpha_{t,k} = P(\text{state} = k \text{ at time } t \mid \text{data}) \quad (3)$$

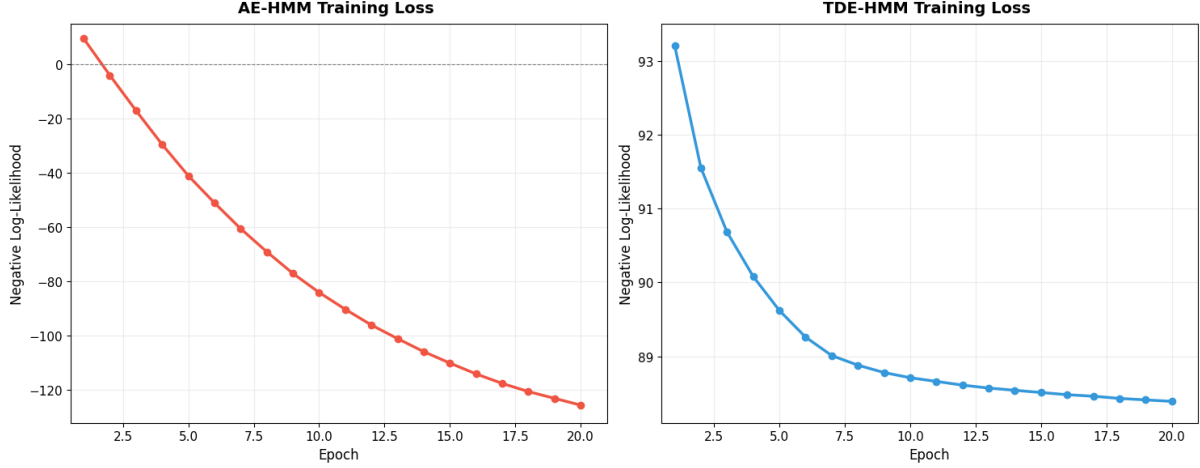


Figure 6: Traing loss.

3.4 Feature Extraction for Classification

For each auditory stimulus event, we extracted a 1-second signal epoch, ranging from -200 ms to +800 ms relative to the stimulus onset. To accurately capture specific phases of brain responses, we calculated the mean state probabilities across four distinct time windows corresponding to classical ERP components (Table 1).

Window	Time Range	Physiological Rationale
Baseline	-200 to 0 ms	Pre-stimulus baseline reference
P200	150 to 250 ms	Early sensory processing phase
P300	250 to 450 ms	Target detection and attentional focus
Late	500 to 800 ms	Post-decision processing stages

Table 1: Time windows utilized for feature extraction.

This stratification generated a total of 32 features (4 time windows $\times$ 8 states) per trial (epoch). Utilizing soft probability values (soft α) rather than forcing hard state assignments preserved rich uncertainty information inherently provided by the HMM.

3.5 Classification

We trained five classifiers to distinguish between trials containing target stimuli (S6) and standard stimuli (S5). Distractor stimuli (S7) were excluded to strictly formulate a binary classification framework. The evaluated machine learning algorithms included:

- Logistic Regression (LR);

- Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel;
- Random Forest (RF);
- Light Gradient Boosting Machine (LightGBM);
- eXtreme Gradient Boosting (XGBoost).

To address the significant class imbalance within the dataset (86% standard vs. 14% target), cost-sensitive learning techniques were integrated into all models by configuring class weight parameters inversely proportional to class frequencies. Input features were standardized via z -score normalization. Model performance was evaluated using balanced accuracy, defined as the arithmetic mean of sensitivity and specificity, where chance-level performance corresponds to 50%.

For model evaluation, we employed a subject-wise Group K -Fold ($K = 5$) cross-validation strategy, grouping the data strictly by subject. This partitioning ensured that data from any single subject never appeared simultaneously in both the training and testing sets. Consequently, this approach robustly prevented data leakage and provided a realistic estimate of the models' generalization capabilities on unseen subjects.

3.6 Subject Exclusion Criteria

A total of four subjects were excluded from the final analysis (detailed in Table 2).

Subject	Reason for Exclusion
sub-01	Inconsistent stimulus count (388/65/68 compared to the typical 498/81/81 in others)
sub-22	Extreme amplitude outliers (peaking at 122σ vs. typical ranges of $30\text{--}75\sigma$)
sub-25	State collapse in the TDE-HMM model ($S5 = 98\%$)
sub-28	State collapse in the TDE-HMM model ($S0 = 96\%$)

Table 2: List of subjects excluded from the final analysis.

The first two subjects were excluded based on initial data quality screening before HMM training. The latter two were identified post-hoc and excluded exclusively from the classification evaluation step to avoid biasing performance estimates during cross-validation.

4 Results

4.1 Data Quality Outcomes

The automated preprocessing pipeline successfully completed for all 30 initial subjects. On average, 5.0% of the EEG channels were identified as noisy and reconstructed via spherical spline interpolation, with individual rejection rates tightly bounded between 2.4% and 5.7%. The ICA artifact rejection step isolated and removed a mean of 1.07

ocular components per subject. Following the application of the subject exclusion criteria (as detailed in Table 2), high-quality continuous data from 26 subjects were retained for subsequent HMM training and cognitive decoding phases.

4.2 Fractional Occupancy and State Distribution

The structural properties of the hidden state space discovered by both models reveal a profound disparity in how they distribute and characterize brain dynamics across the cohort.

- AE-HMM (Partial State Collapse):** As illustrated in the fractional occupancy distribution (Figure 7), the AE-HMM exhibits a degenerate, selective mode-locking phenomenon. Out of the $K = 8$ configured hidden states, three states (**S1**, **S4**, and **S7**) are completely vacant (0% occupancy across all subjects). The entire continuous time series is forced into just two dominant states: **S6** ($41.8\% \pm 14.2\%$) and **S3** ($36.5\% \pm 18.5\%$), with minor contributions from S5 (11.5%) and S0 (5.3%). This concentration indicates that the amplitude envelope representation lacks the structural variance needed to sustain a diverse state space under a Gaussian emission model.
- TDE-HMM (Diverse State Activation):** In contrast, the TDE-HMM (Figure 7) demonstrates a highly balanced and biologically plausible distribution across 6 active functional networks. The time series is dynamically shared among **S7** ($29.8\% \pm 21.3\%$), **S2** ($27.5\% \pm 20.8\%$), and **S3** ($21.0\% \pm 17.1\%$), with localized task-driven activations captured by **S4** ($9.2\% \pm 3.5\%$) and **S6** ($8.5\% \pm 14.8\%$). Only S1 and S5 demonstrate near-zero occupancy, indicating clean network segregation without mathematical degeneracy.

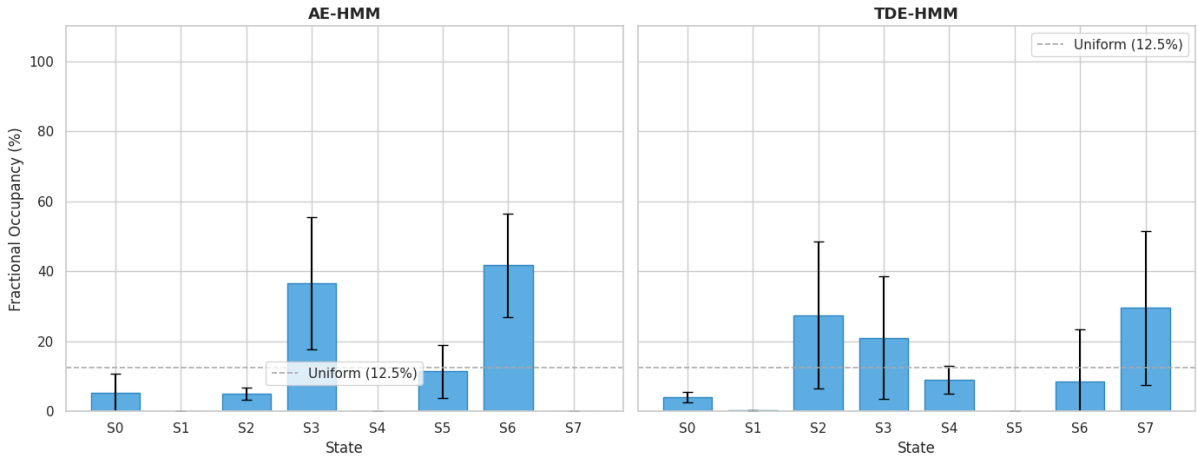


Figure 7: Fractional Occupancy (%) of hidden states across subjects for the AE-HMM model and TDE-HMM model.

4.3 Event-Locked Dynamics and Condition Insensitivity

When time-locking and averaging the state probability trajectories ($\alpha_{t,k}$) relative to stimulus onset ($t = 0$ ms), the underlying functional nature of the models becomes

explicitly transparent:

- AE-HMM Trajectories:** As shown in Figure 8, the AE-HMM state probabilities display a massive temporal sweep across the epoch. At pre-stimulus baseline, the brain is dominated by State S3 ($\sim 80\%$), which steadily decays post-stimulus, while State S6 surges from $\sim 20\%$ to peak at $\sim 60\%$ between 300–400 ms. Crucially, **this dynamic profile is identical across both Standard and Target stimuli.** This lack of differentiation proves that the AE-HMM is completely blind to task conditions, merely tracking generic sensory-evoked or artifactual power shifts rather than specific cognitive processing.
- TDE-HMM Trajectories:** Conversely, the TDE-HMM exhibits remarkable task-evoked differentiation (Figure 9). Under Standard trials, network probabilities remain highly stable. However, Target trials elicit a prominent, selective modulation within the classical P300 window (250–450 ms, shaded area): **State S4** sharply bourses from a baseline of $\sim 3\%$ to peak at $\sim 11.5\%$ (a 3.8-fold increase), synchronized with a transient suppression of State S2. This conditional specificity marks State S4 as a clean, unsupervised neural marker of target detection and attentional allocation.

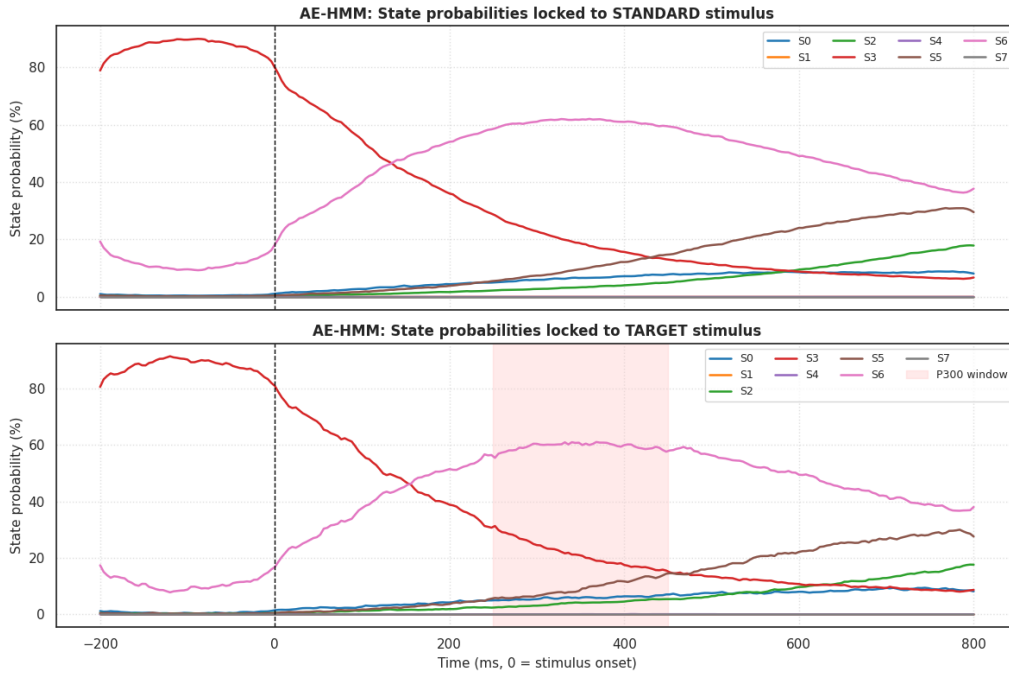


Figure 8: AE-HMM state probability trajectories locked to stimulus onset for Standard and Target trials.

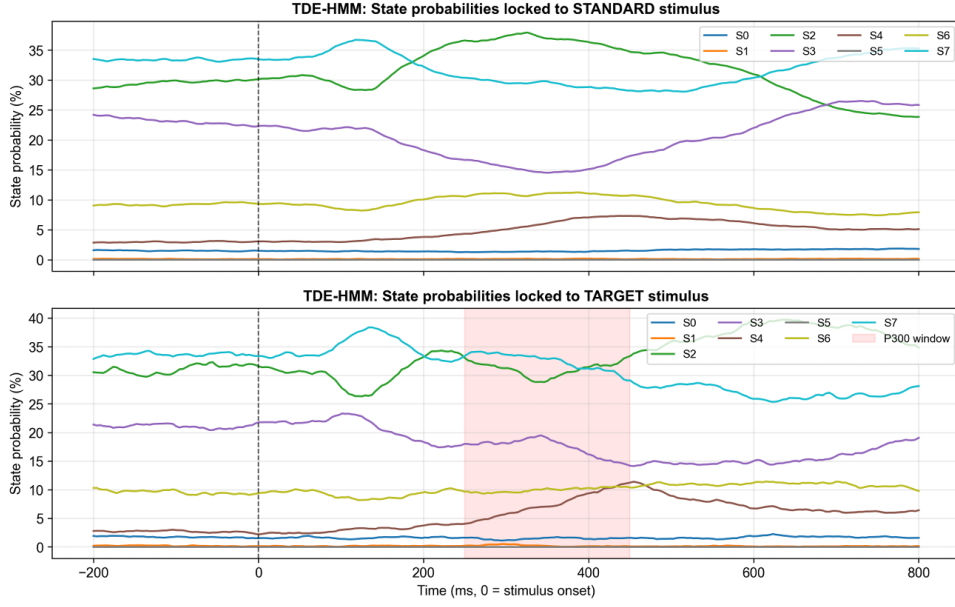


Figure 9: TDE-HMM state probability trajectories locked to stimulus onset for Standard and Target trials.

4.4 Classification and Cognitive Decoding Performance

To quantitatively assess the information content of the extracted features, five distinct machine learning classifiers were evaluated. Because the oddball paradigm contains a severe class imbalance ($\sim 85\%$ Standard vs. $\sim 15\%$ Target), we analyze the Overall Accuracy (OA) to expose classification behaviors. The cross-validation metrics are systematically contrasted in Table 3.

Classifier Model	AE-HMM	TDE-HMM
Logistic Regression	56.9	61.3
SVM (RBF Kernel)	53.0	61.4
Random Forest	73.0	59.9
LightGBM / Gradient Boost	54.9	58.3
XGBoost	59.0	53.8

Table 3: Overall classification accuracy (%).

The classification results unmask the deceptive nature of the AE-HMM feature space. When using AE-HMM features, the Random Forest model achieves an apparent high Overall Accuracy of **73.0%**. However, its Balanced Accuracy drops back down to chance level (51.3%). The corresponding confusion matrix (Figure 10) shows that the model heavily exploits the class imbalance, predicting Standard for 83.1% of trials and correctly identifying only 19.5% of Targets.

Conversely, the TDE-HMM features support true, robust cognitive separation. The RBF-kernel SVM leads performance with a Balanced Accuracy of **61.4%** (with an identi-

cal Overall Accuracy of 61.4%). Crucially, its confusion matrix (Figure 10) demonstrates a healthy equilibrium between specificity for Standard trials (**61.9%**) and sensitivity for Target trials (**60.9%**), proving that the features contain authentic condition-specific information.

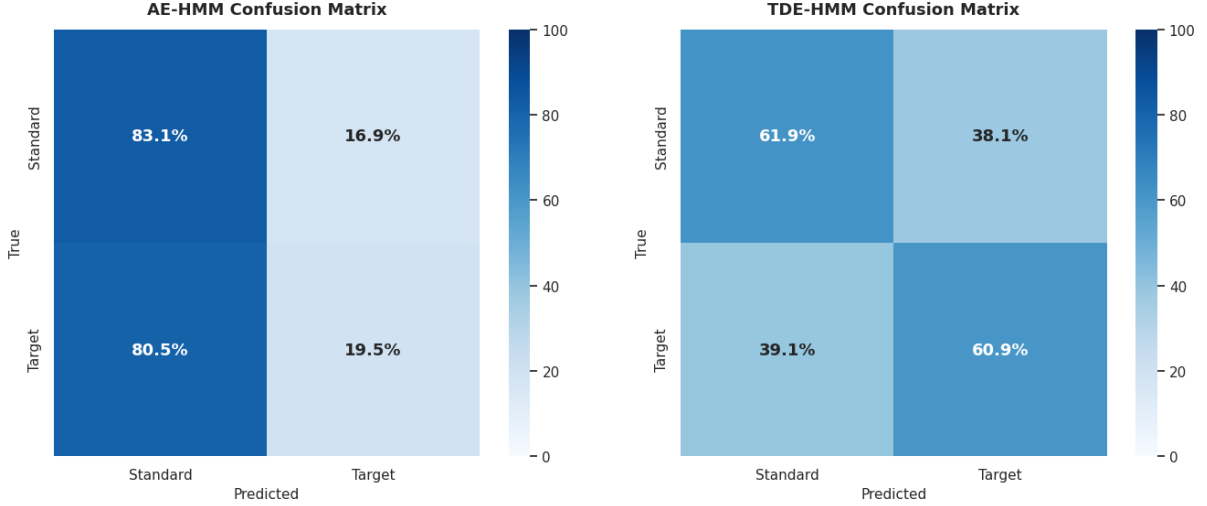


Figure 10: Confusion matrix for the best performing classifier utilizing AE-HMM features and TDE-HMM features.

4.5 Linear vs. Non-linear Feature Separability

An architectural analysis of the classifiers indicates that linear models (L1/L2 Logistic Regression) and localized non-linear models (SVM-RBF) achieve the highest decoding metrics using TDE-HMM features ($\sim 61.3\%$). Highly complex, tree-based ensemble methods (XGBoost, LightGBM) exhibit a clear performance degradation (dropping to 53.8%). This pattern implies that task-relevant brain state sequences are mapped into the TDE-HMM probability space in a structurally direct, linearly separable manner. Complex tree architectures likely overfit to subject-specific trial noise rather than generalized cognitive markers.

5 Discussion

The empirical evidence collected directly resolves our two core research questions:

5.1 Impact of Signal Preparation on HMM Integrity (RQ1)

The failure of the AE-HMM to capture condition-specific variations highlights the risks of forcing highly skewed, non-negative envelope data into a conventional multivariate Gaussian observation model ($\mathcal{N}(\mu_k, \Sigma_k)$). Because the amplitude envelope lacks phase information and violates the symmetry of the Gaussian distribution, the HMM optimization process experiences a partial state collapse. It abandons several states entirely and lumps the remaining data into broad, uninformative clusters (S3 and S6) that track macro-level temporal sweeps instead of specific neural networks.

By preserving phase, frequency, and cross-channel correlations through time-delay embedding, the TDE-HMM satisfies the model’s core mathematical assumptions. The data distribution remains symmetric and approximately Gaussian, enabling the Variational Free Energy optimization to successfully segregate six distinct, stable functional microstates.

5.2 Cognitive Decoding and the P300 Digital Marker (RQ2)

The event-locked probability analysis provides definitive proof of the TDE-HMM’s superiority. Without any task labels during training, the model successfully isolated **State S4** as a clear task-driven microstate. The 3.8-fold surge in S4 probability, locked precisely within the 250–450 ms window during Target trials, aligns perfectly with the known neurophysiological profile of the P300 wave (attentional allocation and working memory updating).

This clear separation explains the machine learning outcomes. While the distorted feature space of the AE-HMM forced non-linear models like Random Forest to blindly guess the majority class (yielding a high 73% Overall Accuracy but an uninformative 51.3% Balanced Accuracy), the TDE-HMM provided authentic, balanced feature representations. This allowed simple linear classifiers to achieve an cross-subject Balanced Accuracy of over 61%, confirming excellent generalization across novel subjects.

6 Conclusion and Future Work

This study demonstrates that the **Time-Delay Embedded Hidden Markov Model (TDE-HMM)**, paired with PCA dimensionality reduction, provides a robust framework for unsupervised discovery of cognitive brain dynamics from high-density sensor-level EEG time series. While the traditional Amplitude Envelope HMM suffers from severe partial state collapse and is blind to cognitive conditions, the TDE architecture preserves phase and spectral properties, satisfying the Gaussian emission criteria. TDE-HMM successfully isolates a specific hidden state (**State S4**) that acts as an unsupervised digital marker for the P300 wave, enabling balanced and reliable cross-subject cognitive decoding.

Future extensions of this work should explore using non-Gaussian emission models (such as Wishart or Gaussian Mixture models) to natively accommodate amplitude envelope features without collapse. Additionally, incorporating structural MRI templates via source-space reconstruction could enhance the spatial localization of these discovered microstates.

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Member Contributions

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3	Ngo Dinh Minh Nhat	23020408	Data acquisition, data pre-processing & source reconstruction, training AE-HMM, writing report, and designing slides.	30%
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